GE M01 Language Fundamentals and Tools

Points: 25 (See Rubric in Canvas)
Due Date: Due date listed in Canvas but some sections will be due as class participation before the final due date. Create Calendar reminders

- Final submission will be accepted up to 24 hours after the due date with a 10% penalty. Meaning if you turn it in at 12:01 am of the next day you will be deducted 10% of the total points from your score.
- If the assignment is more than 24 hours late, it will be a 0.

Submission: Upload files separately and do not upload a zip file.

- **Two Document Files as PDF or Word document:** This document with your answers in the highlighted areas unless otherwise stated and Technical Document
- **Two Java Files (. java files NOT CLASS FILES):** Code from
  - 2.1 Explore GuessNumber Code
  - 2.2 Calculate Grades

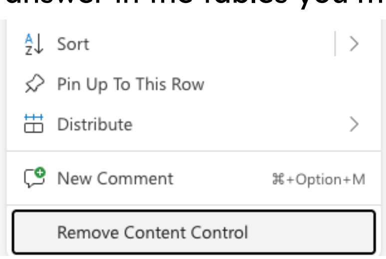
Objectives:

- Analyze and apply module 01 concepts
- Develop effective technical documentation to demonstrate your understanding of technical concepts, with a focus on clarity and professional presentation.
- Apply version control and debugging practices using Eclipse and GitHub.

Effort: You are encouraged to collaborate to discuss concepts and explore writing code together. Remember

- Review lectures and links in the lectures to information first.
- Follow CS Academic Integrity and AI Policy - Harding
- If you find you don't understand the information in the lecture reach out to **Deb, Elaine or Heriberto**

If you download this as a word document and you have any problems typing your answer in the tables you must click on 3 dots and remove content control.



Part 0 Learning and Collaborating Reflection

The reality of computer science is that there are so many languages, technologies and methodologies available and it is constantly evolving with new ones. So the goal is not just to understand the current technologies but developing skills to learn any new technology.

0.1 Working on Teams

In this course, we'll work together on in-class activities, labs, and discussions. Good collaboration isn't just about dividing work — it's about building understanding together. Read [10 Collaboration Skills Examples] and answer the following

1. Pick two of the 10 skills that you think are strengths you bring to a team and why. Two of the 10 skills I see as strengths I cna bring to a team include Active listening and giving and receiving feedback. My current research position has provided me the opportunity to really develop and strengthen these two skills. What I have seen myself develop in these skills through this experience is being able to use active listening to track what content is most beneficial and important when taking notes for a group, and being able to use active listening to give constructive and concise feedback. I also think this experience has provided me a lot of practice in giving and receiving academic feedback. I really appreciate feedback and feel like I do very well at implementing feedback, but I struggled heavily to give feedback to others until being able to practice it constantly through this position.
2 Pick two of the 10 skills you think are most difficult for you and something you can do to try and improve in that area this semester. Two skills that are most difficult for me include conflict resolution, and I still struggle at times to give constructive feedback. When I am in spaces with

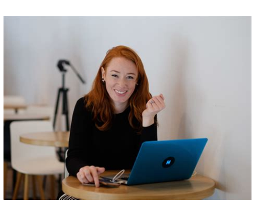
folks who have more experience in a subject, it can be hard for me to branch out or challenge a way of thinking or approach to a topic, especially if other members of a group push back on new approaches to a topic. However, I am hopeful that collaborating in groups often will help me build my confidence in a team. I also hope being able to gain multiple perspectives of how my classmates approach a project will help me to learn how to tailor my feedback to each team-member's learning style so that it can be as beneficial and supportive as possible.

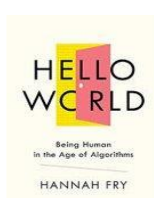
0.2 Learning How to Learn

Learning how to learn is an important component of this course as your career will be full of learning something new. One of my favorite websites is Train Ugly - How To Get More Out Of Your Practice.

1 Listen to this 30 minute podcast Desirable Difficulties - The Learner Lab and answer the following. Explain desirable difficulties and how you think it relates to the learning in this class and in your future career.
2 Read Bluesky CEO Jay Graber warns: If you're a student, using AI means where Bluesky CEO's Jay Graber warns that over-reliance on AI can impair critical thinking and long-term skill development and may contribute to "academic obsolescence" - declining ability to perform without AI What skills do you risk losing if you depend on AI too much?
3 Summarize 3 Strategies you will use this semester to build skills, become a lifelong learner, and Use AI Tools in a way that supports your own thinking.

0.3 Algorithms





"You can't just build an algorithm, put it on a shelf and decide whether you think it's good or bad completely in isolation. You have to think about how

that algorithm actually integrates with the world that you're embedding in."
Hannah Fry

Read Chapter 1 from Hannah Fry, "Hello World, Being Human in the Age of Algorithms" and answer the following.

Answer the following in your own words.

1. Describe an algorithm.
2. Summarize the four main categories of real world tasks that algorithms are doing.
3. Explain the two main approaches taken by algorithms. Include the pros and cons.
4. What can happen when we only rely on machines to make decisions for us?

Part 1: Technical Document Explore and Explain

Purpose: Professional communication to demonstrate your learnings by connecting different ideas from lectures or provided resources.

- Clearly explain key ideas in your own words and include supporting evidence such as code examples, images, screenshots, or charts.

You will update your technical document to have it as a resource for your projects, quizzes, future reference and resume.

- Tech Doc CS1050: Download as word or create a copy of the google documentation.
    - Put in your git repository after you set up git and github.
    - Submit this as a word or pdf with your GE submission.
- Use headings to organize topics and create or update a table of contents to be able to quickly access information.
    - Headings, Subheadings, and Table of Contents (Google Docs)
    - Headings, Subheadings, and Table of Contents (Microsoft Word)
- Complete these sections
    - Module 1: Programming Fundamentals and Tools all objectives
    - Version Control
    - Implement Quality Documented Code

- Apply Java naming conventions for variables, constants, methods, and classes.
- Write comments at top of file and inline comments to document code.

Part 2 Analyze and Apply

Demonstrates a clear analysis of a problem and applies relevant concepts effectively. Work shows independent thinking and understanding. You will use Git, GitHub and Eclipse to edit, version and back up your code.

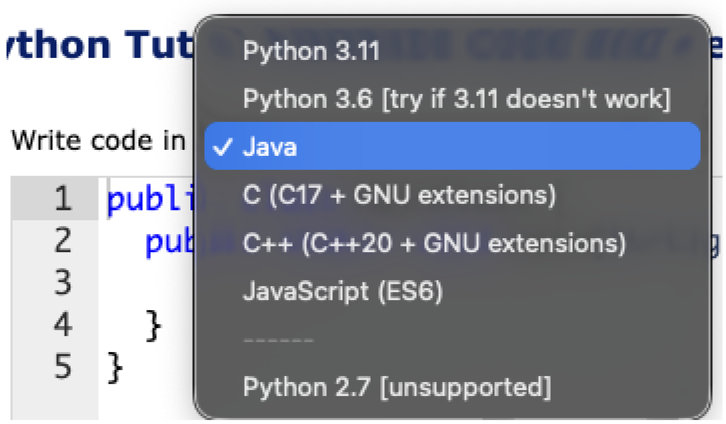
Resources to get started with Git/GitHub and Eclipse

- Git and Github Version Control
- Setup Eclipse - Download and put in your github Resources Folder
- Eclipse Cheat Sheet - Download and put in your github Resources Folder

2.1 Explore GuessNumber Code

Be curious about the code as you explore. Come to office hours if you need help setting up your environment to use Eclipse and Git/Github.

1. Complete Interactive Animation for Guess Number Program
2. Open Eclipse Cheat Sheet and follow the instructions on how to create a project called CS1050M01. Take your time because if you do it wrong it won't work and you will need to do it over.
3. Download GuessNumber.java
4. Open Eclipse Cheat Sheet and read how to add an existing Java file by dragging and dropping into your CS1050M01 Eclipse project.
    a. Use Visualize code in Python, JavaScript, C, C++, and Java if you do not have your environment set up. Select java and paste code so you can edit, run, add comments and step through code.



5. Add comments to the parts of the code to help you understand. You can use parts of the code and comments in your technical document.

6. Version your guess number and technical document locally and then back up on your github cloud server.
7. Upload the .java file into GE M01 submission.

1. Take a screenshot showing you are using the debugger to show what is stored in the variable number and guess after you enter a guess.
2. Describe 3 things you understand better about the code after using the debugger. • • •
3. Identify the warning in the code. Show a screenshot of the warning in the program. Find a resource to help fix the problem based on the warning message. You can use AI for help with syntax errors and warnings.
4. Take a screenshot to show your GuessNumber.java file in your repo on the Github Cloud Server. Explain why only the java file is showing in the repos and not the class file.

2.2 Calculate Grades

You will create a project to store your assignment code and then create your program file. The program you create here will be updated in future guided explorations.

Remember: Commit and push your code often.

1. Use the [Eclipse Cheat Sheet]() to create a new class called **GEM01GradeCalculations** in your CS1050M01 project
2. Update the code to add an initial comment. **This initial comment is required on all assignments.**
    a. Your name
    b. Class name
    c. Section (M/W) or (T/R) Due date
    d. Description

```
/*
 * Name: [YOUR NAME]
 * Class: CS1050 (M/W or T/TH)
 * Description: Guided Exploration 01
 * The program will calculate a final grade for this class based on the category weights
```

```
*/

public class LastNameFirstNameGE01Calculation
{

    {

     }//main

}//Class GE01
```

3. Design your algorithm on paper. Use an example test case to use to determine how to compute a grade on paper.
    a. Use constants for percentage weights and variables for your category grades and final grade. Do not hardcode values in your expression to calculate your final grade. Example of hardcoding (.15 * 95) + (.2* 92) + (.25* 88) + (.2*87) + (.2 *93)
    b. Use correct naming conventions.

Category Weights	Example Category Grades
Class participation 15%	95
Guided Explorations 20%	92
Quizzes 25%	88
Projects 20%	87
Final Demonstration 20%	93

4. **Working with Variables and Expressions:** Add code to calculate your final grade.
    a. Write a little code and run. Do not try and write all the code at once as this is harder to fix errors.
5. Display your name and final grade using **System.out.println**
6. Add comments to explain your code.
7. Commit your .java code file locally and push your code to github remote repository. Make sure you include comments.
8. Upload the .java file into GE M01 submission.

3 Reflection

1. Identify one specific challenge you faced in this exploration and what you did to overcome it.

2. Team Collaboration
What is one strength of yours when working with your group in class?

What is one strength of your team?

What is one area you would like to improve about your team working together in class?

3. Environment Set Up

On a scale of 1 to 5 rate yourself on using the following tools and include one thing you understand and one thing you need help on.

git/github rating:

- I understand how to
- I need help on

If you are not using Eclipse include what IDE you are using
Eclipse rating:

- I understand how to
- I need help on

4. Now that you are submitting your first Guided Exploration

- What do you wish you would have asked for help on or what did you get help on?
- How was your time management? Did you have to do a lot the day it was due?
- What will you keep the same and what will you try to improve on when you do the next GE?